

Errors

- Programs can contain errors at different levels.
- Programs can be:
 1. **Lexical errors:** such as misspelling of identifier, keyword or operator.
 2. **Syntactic errors:** such as arithmetic expressions with unbalanced parenthesis.
 3. **Semantic errors:** such as an operator applied to an incompatible operand.
 4. **Logical errors:** such as infinitely recursive call due to transfer of control to a block that does not exist.

Top-Down Parsing

- It can be viewed as an attempt to construct a parse tree for the input starting from the root and creating the nodes of the parse tree in pre-order.

Recursive Descent Parsing

- It is a top-down method of syntax analysis in which we execute a set of recursive procedures to process the input.
- Predictive parsing is a special form of recursive descent parsing in which the lookahead symbol unambiguously determines the procedure selected for each non-terminal.

Functions for Parsing

1. FIRST

- If α is any string of grammar symbols, then $FIRST(\alpha)$ is the set of terminals that begins the strings derived from α , i.e. the set of terminals that can appear at the far left of any parse tree derived from a non-terminal.
- If $\alpha \Rightarrow^* \epsilon$, then $\epsilon \in FIRST(\alpha)$.
- Rules for finding $FIRST(X)$:
 1. If X is a terminal, then $FIRST(X)$ is $\{X\}$.
 2. If X is a non-terminal and there is a production of the form $X \rightarrow Y_1 Y_2 \dots Y_k$, then place a in $FIRST(X)$ if for some i , a is in $FIRST(Y_i)$, and ϵ is in all of $FIRST(Y_1), FIRST(Y_2), \dots, FIRST(Y_{i-1})$. If ϵ is in all of $FIRST(Y_i)$ from 1 to k , then also add ϵ to $FIRST(X)$.
 3. If $A \rightarrow \epsilon$, then add ϵ to $FIRST(A)$.

2. FOLLOW

- We define $FOLLOW(A)$ for non-terminal A to be the set of terminals a that can appear immediately to the right of A in some sentential form, i.e. the set of terminals a such that there exists a derivation of the form $S \Rightarrow^* \alpha A a \beta$ for some α and β .
- Rules for finding FOLLOW:
 1. Place $\$$ in $FOLLOW(S)$ where S is the start symbol and $\$$ is the input end marker.
 2. If there is a production, $A \rightarrow \alpha B \beta$, then everything in $FIRST(\beta)$ except for ϵ is placed in $FOLLOW(B)$.
 3. If there is a production $A \rightarrow \alpha B$ or a production $A \rightarrow \alpha B \beta$, where $FIRST(\beta)$ contains ϵ , i.e. $\beta \Rightarrow^* \epsilon$, then everything in $FOLLOW(A)$ is in $FOLLOW(B)$.
- Little modification (makes it easier to implement):
 1. Add $\$$ to $FOLLOW(S)$.
 2. For each production rule of the form $A \rightarrow B_1 B_2 \dots B_k$
 1. Add everything from $FOLLOW(A)$ to $FOLLOW(B_k)$.
 1. For each $B_i \in V \cup T$ such that $i = 1, 2, \dots, k-1$
 1. Add everything from $FIRST(B_{i+1} B_{i+2} \dots B_k)$ except ϵ to $FOLLOW(B_i)$.
 2. If $FIRST(B_{i+1} B_{i+2} \dots B_k)$ contains ϵ , add everything from $FOLLOW(A)$ to $FOLLOW(B_i)$.
- Note that add everything from FOLLOW(X) to FOLLOW(Y) operations are to be deferred till the end because X may not be completely filled at the stage it was called (best to keep a reference to FOLLOW(X) and not the actual values).

Example 1

Consider a grammar with the following production rules:

1. $E \rightarrow TE'$
2. $E' \rightarrow +TE' \mid \epsilon$
3. $T \rightarrow FT'$
4. $T' \rightarrow *FT' \mid \epsilon$
5. $F \rightarrow (E) \mid id$

$$FIRST(E) = FIRST(T) = FIRST(F) = \{ (, id \}$$

$$FIRST(E') = \{ +, \epsilon \}$$

$$FIRST(T') = \{ *, \epsilon \}$$

Finding FOLLOW:

- Add $\$$ to $FOLLOW(E)$.
- From production 1:
 - Add everything from $FOLLOW(E)$ to $FOLLOW(E')$.
 - Add everything from $FIRST(E')$ to $FOLLOW(T)$.

- Since $FIRST(E')$ contains ϵ , add everything from $FOLLOW(E)$ to $FOLLOW(T)$.
- From production 2:
 - Add everything from $FOLLOW(E')$ to $FOLLOW(T)$.
- From production 3:
 - Add everything from $FOLLOW(T)$ to $FOLLOW(T')$.
 - Add everything from $FIRST(T')$ to $FOLLOW(F)$.
 - Since $FIRST(T')$ contains ϵ , add everything from $FOLLOW(T)$ to $FOLLOW(F)$.
- From production 4:
 - Add everything from $FOLLOW(T')$ to $FOLLOW(F)$.
- From production 5:
 - Add) to $FOLLOW(E)$.

$$FOLLOW(E) = FOLLOW(E') = \{\$, \text{)}\}$$

$$FOLLOW(T) = FOLLOW(T') = \{+, \$, \text{)}\}$$

$$FOLLOW(F) = \{*, +, \$, \text{)}\}$$

Example 2

Consider a grammar with the following production rules:

1. $S \rightarrow iEtSS' \mid a$
2. $S' \rightarrow eS \mid \epsilon$
3. $E \rightarrow b$

$$FIRST(S) = \{i, a\}$$

$$FIRST(S') = \{e, \epsilon\}$$

$$FIRST(E) = \{b\}$$

Finding FOLLOW:

- Add \$ to $FOLLOW(S)$.
- From production 1:
 - Add everything from $FOLLOW(S)$ to $FOLLOW(S')$.
 - Add t to $FOLLOW(E)$.
 - Add everything from $FIRST(S')$ to $FOLLOW(S)$.
- From production 2:
 - Add everything from $FOLLOW(S')$ to $FOLLOW(S)$.

$$FOLLOW(S) = \{\$, e\}$$

$$FOLLOW(S') = \{\$, e\}$$

$$FOLLOW(E) = \{t\}$$

Example 3

Consider a grammar with the following production rules:

1. $E \rightarrow TE'$
2. $E' \rightarrow +E \mid \epsilon$
3. $T \rightarrow FT'$
4. $T' \rightarrow T \mid \epsilon$
5. $F \rightarrow PF'$
6. $F' \rightarrow *F' \mid \epsilon$
7. $P \rightarrow (E) \mid a \mid b \mid \epsilon$

$$FIRST(E) = FIRST(T) = FIRST(F) = FIRST(P) \cup FIRST(F') = \{ (, a, b, *, \epsilon \}$$

$$FIRST(E') = \{ +, \epsilon \}$$

$$FIRST(T') = FIRST(T) = \{ (, a, b, *, \epsilon \}$$

$$FIRST(P) = \{ (, a, b, \epsilon \}$$

$$FIRST(F') = \{ *, \epsilon \}$$

Finding FOLLOW:

- Add \$ to $FOLLOW(E)$.
- From production 1:
 - Add everything from $FOLLOW(E)$ to $FOLLOW(E')$.
 - Add everything from $FIRST(E')$ to $FOLLOW(T)$.
 - Since $FIRST(E')$ contains ϵ , add everything from $FOLLOW(E)$ to $FOLLOW(T)$.
- From production 2:
 - Add everything from $FOLLOW(E')$ to $FOLLOW(E)$.
- From production 3:
 - Add everything from $FOLLOW(T)$ to $FOLLOW(T')$.
 - Add everything from $FIRST(T')$ to $FOLLOW(F)$.
 - Since $FIRST(T')$ contains ϵ , add everything from $FOLLOW(T)$ to $FOLLOW(F)$.
- From production 4:
 - Add everything from $FOLLOW(T')$ to $FOLLOW(T)$.
- From production 5:
 - Add everything from $FOLLOW(F)$ to $FOLLOW(F')$.
 - Add everything from $FIRST(F')$ to $FOLLOW(P)$.
 - Since $FIRST(F')$ contains ϵ , add $FOLLOW(F)$ to $FOLLOW(P)$.
- From production 7:
 - Add) to $FOLLOW(E)$.

$$FOLLOW(E) = FOLLOW(E') = \{ \$,) \}$$

$$FOLLOW(T) = FOLLOW(T') = \{ +, \$,) \}$$

$$FOLLOW(F) = FOLLOW(F') = FOLLOW(P) = \{ (, a, b, *, +, \$,) \}$$